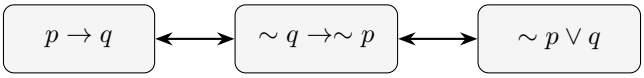
DISCRETE STRUCTURES

CMSC250

Chapter 3: Implications and Logical Equivalence

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Conditionals, biconditionals, logical laws, and rigorous derivations



Notation. Negation is written $\sim p$ throughout these notes. The alternative symbol $\neg p$ has the same meaning.

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1 Why implications matter

Mathematics is full of claims of the form "if a condition holds, then a conclusion follows." For instance, "If an integer is divisible by 4, then it is even" does not assert that every integer is divisible by 4. Instead, it promises that *whenever* the first condition is met, the second condition must be met as well.

We write the conditional statement as $p \to q$, read "if p, then q." Here p is the **hypothesis** (also called antecedent) and q is the **conclusion** (also called consequent). An implication rules out exactly one situation: the hypothesis holds but the conclusion fails.

2 The truth table for implication

p	q	$p \to q$	Explanation
1	1	1	The promised conclusion holds.
1	0	0	The promise is broken.
0	1	1	The hypothesis never occurs.
0	0	1	The hypothesis never occurs.

Key idea. A conditional $p \to q$ is **false only when** $p = 1$ **and** $q = 0$. In every other interpretation it is true.

2.1 Why is a conditional true when its hypothesis is false?

Consider the rule "If you score 90 or more, you receive an A." Someone who scores 85 and receives a B has not violated the rule; the rule made no promise about that student. Similarly, a student scoring 85 who receives an A does not violate the rule. The only violation would be scoring at least 90 but not receiving an A. In propositional logic, a conditional with a false hypothesis is called **vacuously true**.

This does *not* mean that the hypothesis is secretly true, or that the conditional claims a cause-and-effect relationship. It means only that the conditional has not been contradicted.

Worked example: evaluate a conditional

Let $p = 0$ and $q = 0$. Then $p \to q = 1$. By contrast, if $p = 1$ and $q = 0$, the conditional is 0. To evaluate a complicated conditional, evaluate its hypothesis and conclusion separately first.

3 Equivalent ways to express an implication

A crucial equivalence is

$$\boxed{p \to q \ \equiv \sim p \vee q.}$$

Why? The right side is false only when $\sim p = 0$ and $q = 0$, which means $p = 1, q = 0$. That is precisely the false row of the implication.

p	q	$p \to q$	$\sim p$	$\sim p \vee q$
1	1	1	0	1
1	0	0	0	0
0	1	1	1	1
0	0	1	1	1

Because the two final expressions match for every interpretation, they are **logically equivalent**, denoted $\equiv$.

Another useful form is $p \to q \equiv \sim (p \wedge \sim q)$. It says that there is no interpretation in which p is true and q is false.

4 Converse, inverse, and contrapositive

Given the original conditional $p \to q$, we can construct three related statements:

Name	Formula	How it is formed
Original	$p \to q$	Keep the original hypothesis and conclusion.
Converse	$q \to p$	Swap the hypothesis and conclusion.
Inverse	$\sim p \to \sim q$	Negate both, without swapping.
Contrapositive	$\sim q \to \sim p$	Swap and negate both.

Worked example: the divisibility example

Let p mean "n is divisible by 4" and q mean "n is even."

- Original: If n is divisible by 4, then n is even. **True.**
- Converse: If n is even, then n is divisible by 4. **False**; $n = 6$ is a counterexample.
- Inverse: If n is not divisible by 4, then n is not even. **False**; again $n = 6$.
- Contrapositive: If n is not even, then n is not divisible by 4. **True.**

The original and contrapositive always have the same truth value. The converse and inverse also always have the same truth value, but they need not match the original.

4.1 Proving the contrapositive equivalence

p	q	$p \to q$	$q \to p$	$\sim p \to \sim q$	$\sim q \to \sim p$
1	1	1	1	1	1
1	0	0	1	1	0
0	1	1	0	0	1
0	0	1	1	1	1

Reading the columns shows

$$\boxed{p \to q \equiv \sim q \to \sim p}, \qquad \boxed{q \to p \equiv \sim p \to \sim q}.$$

Key idea. Never assume the converse is true just because the original is true. A counterexample to a converse is often much easier to find than a proof of the original.

5 Necessary and sufficient conditions

The English phrases *necessary* and *sufficient* encode the direction of an implication.

If $p \to q$, then p is **sufficient** for q: knowing p is true guarantees q. Meanwhile q is **necessary** for p: p cannot be true without q also being true.

For example, being divisible by 4 is sufficient for being even. Being even is necessary for being divisible by 4. Neither statement says that being even is sufficient for being divisible by 4.

English phrase	Logical translation
p is sufficient for q	$p \to q$
q is necessary for p	$p \to q$
p only if q	$p \to q$
p if q	$q \to p$
p whenever q	$q \to p$
p is necessary and sufficient for q	$p \leftrightarrow q$

Worked example: the difference between "if" and "only if"

"I will go running *if* it is sunny" means sunny $\to$ running. "I will go running *only if* it is sunny" means running $\to$ sunny. In the second statement, sunshine is a requirement for running, not a guarantee that running will happen.

6 Biconditionals: if and only if

A **biconditional** $p \leftrightarrow q$ means both $p \to q$ and $q \to p$ hold. It is true precisely when p and q have the same truth value.

p	q	$p \to q$	$q \to p$	$p \leftrightarrow q$
1	1	1	1	1
1	0	0	1	0
0	1	1	0	0
0	0	1	1	1

Thus

$$\boxed{p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)}.$$

The phrase *if and only if* is often shortened to *iff*. In a proof of an iff statement, we usually establish the forward implication and then the reverse implication separately.

Worked example: an exact characterization

For an integer n, "n is even iff n^2 is even" asserts two directions: (1) if n is even, then n^2 is even; (2) if n^2 is even, then n is even. Both are true, but a complete proof must justify both directions.

7 Logical equivalence and the standard laws

Two expressions A and B are **logically equivalent** when they have the same truth value under *every* interpretation. We write $A \equiv B$. This is stronger than observing that they happen to agree in one example.

The following laws are tools for replacing one expression with an equivalent expression. Here p, q, r denote arbitrary propositions, $\top$ is a tautology, and $\bot$ is a contradiction.

Law	Equivalences
Commutative	$p \wedge q \equiv q \wedge p$; $p \vee q \equiv q \vee p$
Associative	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$; similarly for $\vee$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
Identity	$p \wedge \top \equiv p$; $p \vee \bot \equiv p$
Universal bounds	$p \vee \top \equiv \top$; $p \wedge \bot \equiv \bot$
Idempotent	$p \wedge p \equiv p$; $p \vee p \equiv p$
Negation	$p \vee \sim p \equiv \top$; $p \wedge \sim p \equiv \bot$
Double negation	$\sim (\sim p) \equiv p$
De Morgan	$\sim (p \wedge q) \equiv \sim p \vee \sim q$
	$\sim (p \vee q) \equiv \sim p \wedge \sim q$
Absorption	$p \vee (p \wedge q) \equiv p$; $p \wedge (p \vee q) \equiv p$
Conditional	$p \to q \equiv \sim p \vee q$
Biconditional	$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$

7.1 Why De Morgan's laws work

The negation of "both p and q" means that at least one of the two fails: $\sim (p \wedge q) \equiv \sim p \vee \sim q$. Likewise, the negation of "at least one holds" requires that neither holds: $\sim (p \vee q) \equiv \sim p \wedge \sim q$. Notice that **the connective changes** when negation moves inside.

8 Writing equivalence proofs one law at a time

In a formal derivation, every line must be equivalent to the previous line, and each step should name the law that permits the replacement. Do not jump from the starting expression to the final answer merely because you recognize the result.

Worked example: derive an implication identity

Prove $\sim (p \to q) \equiv p \wedge \sim q$:

$$\begin{aligned} \sim (p \to q) &\equiv \sim (\sim p \vee q) && \text{Conditional law} \\ &\equiv \sim (\sim p) \wedge \sim q && \text{De Morgan's law} \\ &\equiv p \wedge \sim q && \text{Double negation law.} \end{aligned}$$

This derivation also explains why the only way to make an implication false is to make its hypothesis true and its conclusion false.

Worked example: a longer equivalence proof

Show $(p \to q) \wedge p \equiv p \wedge q$:

$$\begin{aligned} (p \to q) \wedge p &\equiv (\sim p \vee q) \wedge p && \text{Conditional law} \\ &\equiv p \wedge (\sim p \vee q) && \text{Commutative law} \\ &\equiv (p \wedge \sim p) \vee (p \wedge q) && \text{Distributive law} \\ &\equiv \bot \vee (p \wedge q) && \text{Negation law} \\ &\equiv (p \wedge q) \vee \bot && \text{Commutative law} \\ &\equiv p \wedge q && \text{Identity law.} \end{aligned}$$

Each change is small enough to check independently. This style makes it easy to locate an error and satisfies the course expectation that laws be cited explicitly.

9 Truth tables versus algebraic derivations

A **truth table** proves equivalence by checking all possible assignments. For n distinct propositional variables there are 2^n rows. An **algebraic derivation** proves equivalence by applying known laws, one justified step at a time. Either method can establish a general equivalence, but when a problem explicitly requires named laws, a truth table alone does not satisfy that instruction.

To **disprove** equivalence, however, one counterexample interpretation is enough. For example, $p \to q$ and $q \to p$ differ when $p = 1, q = 0$: the first is false and the second is true.

10 Common mistakes and grading pitfalls

1. **Reversing "only if."** "p only if q" means $p \to q$, not $q \to p$.
2. **Treating a false hypothesis as making an implication false.** A conditional is false only in the $1 \to 0$ row.
3. **Confusing converse and contrapositive.** The contrapositive swaps *and* negates; the converse only swaps.
4. **Negating an implication incorrectly.** $\sim (p \to q)$ is $p \wedge \sim q$, not $\sim p \to \sim q$.

5. **Skipping equivalence steps.** Show intermediate expressions, including double negation and commutativity when needed.
6. **Not citing laws.** State the precise rule justifying each line of a symbolic derivation.
7. **Using $=$ in place of $\equiv$.** The equivalence symbol describes agreement of truth values under every interpretation.
8. **Proving only one direction of an iff statement.** Both implications must be justified.

11 Practice problems

Attempt the problems before reading the solutions. For truth tables, use 1 and 0 and descending binary row order.

1. Construct the truth table for $p \to q$. Explain which row makes it false.

2. Let p mean "the server is online" and q mean "the site is accessible." Write the converse, inverse, and contrapositive of $p \to q$ in English.

3. Translate "I submit the assignment only if the tests pass" using $s =$ "I submit" and $t =$ "the tests pass."

4. Translate "Passing the final is sufficient for passing the course" using $f =$ "pass the final" and $c =$ "pass the course."

5. Determine whether $(p \to q)$ and $(\sim q \to \sim p)$ are equivalent using a truth table.

6. Find a truth assignment showing that $p \to q$ and $q \to p$ are not equivalent.

7. Write a biconditional as a conjunction of two implications, then eliminate the implication symbols.

8. Prove $\sim (p \to q) \equiv p \wedge \sim q$ using named laws, one law per line.

9. Prove $(p \vee q) \wedge \sim p \equiv \sim p \wedge q$ using named laws, one law per line.

10. Classify $(p \to q) \vee (q \to p)$ as a tautology, contradiction, or contingency. Justify.

11. Translate "You can enter the lab if and only if you have a badge" into symbols, and state both implications in English.

12. Explain why $p \leftrightarrow q$ is true when $p = q = 0$.

12 Detailed solutions

1.

p	q	$p \to q$
1	1	1
1	0	0
0	1	1
0	0	1

The only false row has a true hypothesis and false conclusion.

2. Converse: If the site is accessible, the server is online. Inverse: If the server is not online, the site is not accessible. Contrapositive: If the site is not accessible, the server is not online. These are translations, not claims that all four are true.

3. $s \to t$. Passing the tests is a necessary condition for submitting the assignment.

4. $f \to c$. The word "sufficient" identifies f as the hypothesis.

5.

p	q	$p \to q$	$\sim q \to \sim p$
1	1	1	1
1	0	0	0
0	1	1	1
0	0	1	1

The columns match, so the formulas are equivalent.

6. Set $p = 1, q = 0$. Then $p \to q = 0$, while $q \to p = 1$. A single disagreement disproves equivalence.

7. $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p) \equiv (\sim p \vee q) \wedge (\sim q \vee p)$. The second step applies the conditional law to each implication; if required to use strictly one law per line, write the two replacements as separate lines.

8.
$$\begin{aligned} \sim (p \to q) &\equiv \sim (\sim p \vee q) && \text{Conditional law} \\ &\equiv \sim (\sim p) \wedge \sim q && \text{De Morgan's law} \\ &\equiv p \wedge \sim q && \text{Double negation law.} \end{aligned}$$

9.
$$\begin{aligned} (p \vee q) \wedge \sim p &\equiv \sim p \wedge (p \vee q) && \text{Commutative law} \\ &\equiv (\sim p \wedge p) \vee (\sim p \wedge q) && \text{Distributive law} \\ &\equiv (p \wedge \sim p) \vee (\sim p \wedge q) && \text{Commutative law} \\ &\equiv \bot \vee (\sim p \wedge q) && \text{Negation law} \\ &\equiv (\sim p \wedge q) \vee \bot && \text{Commutative law} \\ &\equiv \sim p \wedge q && \text{Identity law.} \end{aligned}$$

10. It is a tautology. If $p = 0$, then $p \to q$ is true. If $p = 1$ and $q = 1$, both implications are true. If $p = 1$ and $q = 0$, then $q \to p$ is true. Thus at least one implication is true in every row.

11. Let e = "you can enter the lab" and b = "you have a badge." The biconditional is $e \leftrightarrow b$. Its directions are "If you can enter, then you have a badge" and "If you have a badge, then you can enter."

12. If both are false, $p \to q = 1$ and $q \to p = 1$ because both hypotheses are false. Their conjunction is true, so $p \leftrightarrow q = 1$.